

Investigating Crash Severity in Victoria

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Introduction

- Road safety is a core objective for the Victorian government and the Transport Accident Commission (TAC). Aiming to halve the amount of road deaths by 2030 and have zero road deaths by 2050. (Department of Transport, 2020)
- Despite progress, during 2025, 291 people were killed on Victorian roads, a 2.5% increase. Additionally, only 7 of the 79 Local government areas in Victoria recorded zero deaths within 2025. (Transport Accident Commission, 2026)
- Understanding where and when the most severe crashes occur is important for determining where to target road safety investment.



travelvictoria, n.d, (<https://www.travelvictoria.com.au/victoria/driving/roads/>)

Problem Statement

This report aims to investigate whether:

- Crashes occurring on Country roads result in a higher number of injuries/deaths than accidents on Metro roads.
- Are crashes which occur at night associated with severity of injury when compared to crashing during the day.

Both questions are important to research and can provide useful information in determining strategies for increased road safety

To do this we will use two main statistical tests

- A two-sample t-test will be conducted to test for a significant difference between the mean persons injured or killed in accidents
- A chi-square test will be used to determine categorical association between time of day and the severity of injury.

Data

The data used in this investigation is the **Victoria road crash data**, available from [DATAVIC](#)

This data set contains information about each road crash that has occurred within Victoria since 2012, and is updated on a monthly basis with a 7-month lag.

As of (31 March 2026) it contains 197,249 observations.

For the following investigation, we will be focusing on the following 4 variables:

- **LIGHT_CONDITION**: Categorical value indicating the light condition / level of brightness
- **SEVERITY**: An estimation of the severity of the accident.
- **INJ_OR_FATAL**: The number of people in the accident injured or killed.
- **STAT_DIV_NAME**: Indication of whether the accident occurred in Metro or Country Victoria

```
data <- read_csv("victorian_road_crash_data.csv")
```

```
data %<>% select('LIGHT_CONDITION', 'SEVERITY', 'INJ_OR_FATAL', 'STAT_DIV_NAME')
```

Data preprocessing

- To begin the preprocessing we filter out the four observations containing non injury accidents as they are not within the research question of determining the severity of the injury sustained in crashes.

```
# Filter non injury accidents
data %<>% filter(SEVERITY != 'Non injury accident')
```

- For this investigation we are using the light condition variable as a means to determine day/night. As such, we re-code the labels in the data based on whether it is bright or dark at the time of the crash. Additionally, we set observations with “Unknown” light levels to be NA.

```
# Re-code the light condition into 'Day' and 'Night', leaving Unknown as NA.
data %<>% mutate(LIGHT_CONDITION=dplyr::recode(LIGHT_CONDITION, 'Day' = 'Day',
                                              'Dusk/Dawn' = 'Day',
                                              'Dark Street lights on' = 'Night',
                                              'Dark Street lights off' = 'Night',
                                              'Dark No street lights' = 'Night',
                                              'Dark Street lights unknown' = 'Night',
                                              'Unk.' = NA_character_))
```

- We then convert variables of character data type into factors with their proper labels.

```
# Factor the categorical variables
data$LIGHT_CONDITION %<>% factor(levels=c('Day', 'Night'))
data$STAT_DIV_NAME %<>% factor(levels=c('Metro', 'Country'))
data$SEVERITY %<>% factor(levels=c('Fatal accident', 'Other injury accident', 'Serious injury accident'))
```

- From checking the data, we find that only 3.2% of the observations contain missing values. Thus we will omit them from the analysis.

```
mean(!complete.cases(data))
```

```
## [1] 0.03212249
```

```
# Omit observations with missing values
data %<>% na.omit()
```

Descriptive Statistics and Visualisation

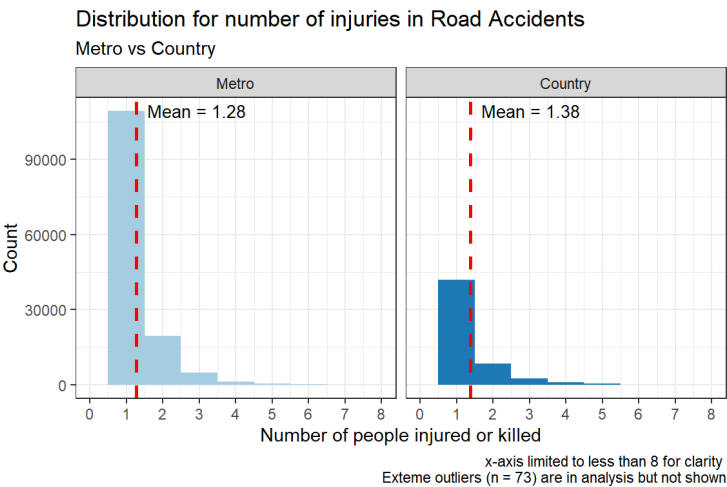
```
data %>% group_by(STAT_DIV_NAME) %>% summarise(Min = min(INJ_OR_FATAL, na.rm = TRUE),
  Q1 = quantile(INJ_OR_FATAL, probs = .25, na.rm = TRUE),
  Median = median(INJ_OR_FATAL, na.rm = TRUE),
  Q3 = quantile(INJ_OR_FATAL, probs = .75, na.rm = TRUE),
  Max = max(INJ_OR_FATAL, na.rm = TRUE),
  Mean = mean(INJ_OR_FATAL, na.rm = TRUE),
  SD = sd(INJ_OR_FATAL, na.rm = TRUE),
  n = n()) -> table1

knitr::kable(table1)
```

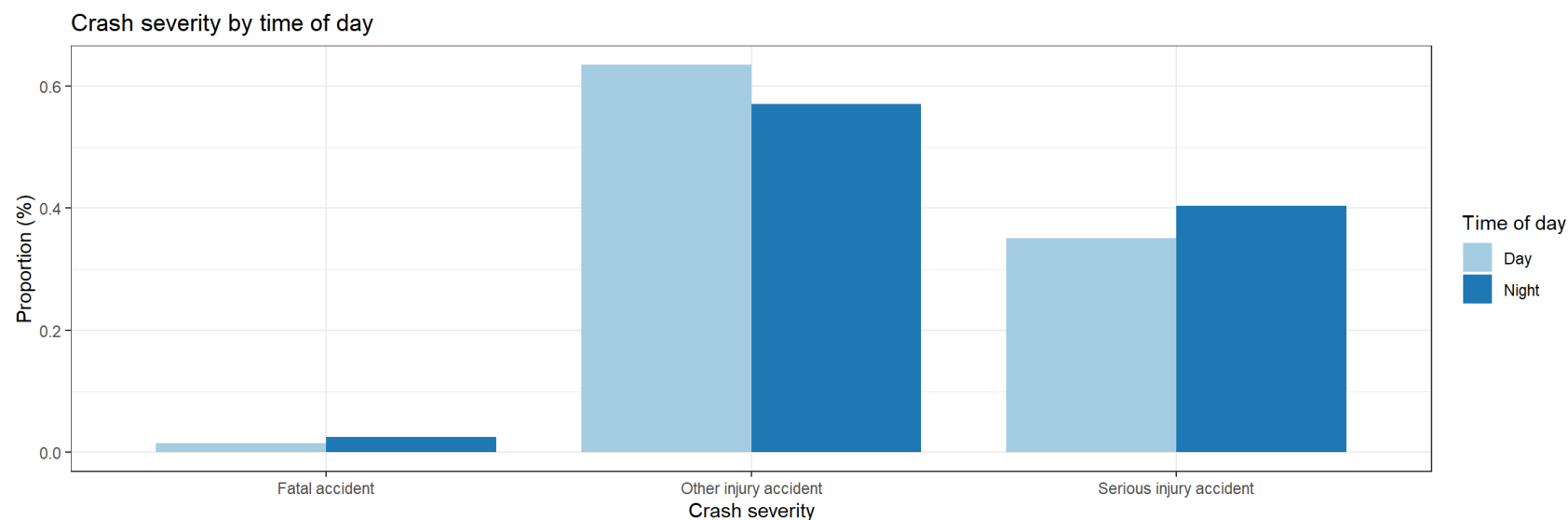
STAT_DIV_NAME	Min	Q1	Median	Q3	Max	Mean	SD	n
Metro	0				46	1.278092	0.6922737	135944
Country	0				48	1.375566	0.9158839	54965

Outliers

From the histogram below we can see that the Number of injuries and death distribution is heavily right-skewed with a small number of high-injury crashes. The x-axis was limited to 8 for clarity, but all the outliers were kept for the analysis as they represent legitimate events and not data errors.



Descriptive Statistics and Visualisation cont.



The figure above shows that crashes that occurred during the night had a proportionally higher rate of fatal and serious injuries compared to the day. This motivated the chi-test conducted further in the report.

Descriptive Statistics and Visualisation (R-Code)

```
# get Means
means <- data %>% group_by(STAT_DIV_NAME) %>%
  summarise(Mean = mean(INJ_OR_FATAL, na.rm = TRUE))

# Histogram plot
ggplot(data, aes(x = INJ_OR_FATAL, fill = STAT_DIV_NAME)) +
  geom_histogram(binwidth = 1) +
  geom_vline(data = means, aes(xintercept = Mean),
    color = 'red',
    linewidth=1,
    linetype='dashed') +
  geom_text(data = means, aes(x= Mean + 0.3, y = Inf,
    label = paste0("Mean = ", round(Mean, 2))),
    hjust = 0,
    vjust = 1.5,
    size = 3.5) +
  facet_wrap(~ STAT_DIV_NAME) +
  coord_cartesian(xlim = c(0, 8))+
  labs(title = 'Distribution for number of injuries in Road Accidents', subtitle = "Metro vs Country",
    x = "Number of people injured or killed", y = "Count",
    caption = "x-axis limited to less than 8 for clarity \nExteme outliers (n = 73) are in analysis but not shown")+
  theme_bw() +
  scale_fill_brewer(palette="Paired") +
  scale_x_continuous(breaks = (0:8)) +
  theme(legend.position="none")
```

```
# Proportion table
P <- table(data$LIGHT_CONDITION, data$SEVERITY) %>% prop.table(margin = 1)

# Bar plot
ggplot(as.data.frame(P), aes(x = Var2, y = Freq, fill = Var1)) +
  geom_bar(stat="identity", position = "dodge", width = 0.8)+
  labs(title = "Crash severity by time of day",
    x = "Crash severity",
    y = "Proportion (%)") +
  theme_bw() +
  scale_fill_brewer(palette="Paired") +
  guides(fill=guide_legend(title="Time of day"))
```


Hypothesis Testing and Confidence Interval

We are going to investigate whether there is a significant difference between the mean number of people injured for accidents occurring in Metro and Country Victoria.

$$H_0 : \mu_1 = \mu_2$$

$$H_A : \mu_1 \neq \mu_2$$

μ_1 is the mean number of injuries of Metro accidents and μ_2 represents accidents occurring in country Victoria.

Assumptions we will check for:

- The populations being compared are independent of each other.
- Both populations are normally distributed.
- Both populations have equal variance.

Assumptions For Hypothesis Testing

Independence

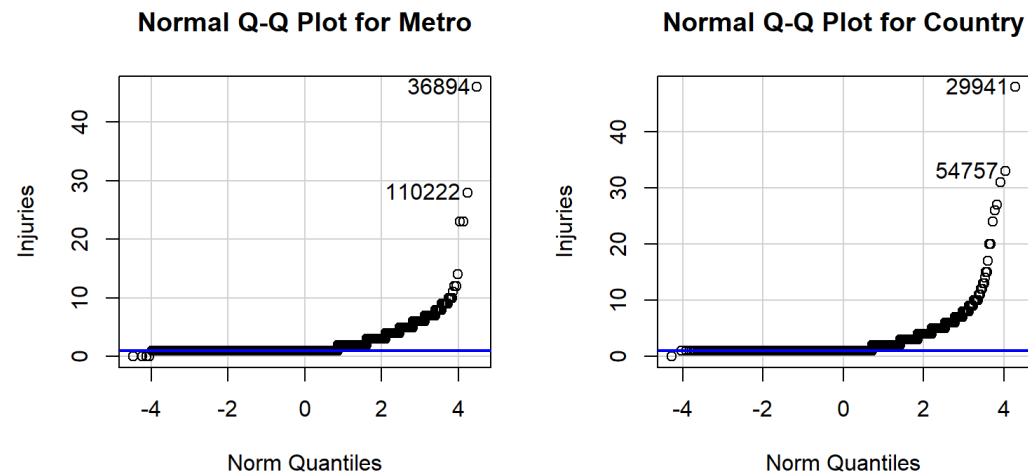
Because each observation in the data is a separate accident event, as such, the first assumption for independence is met.

Normality

To check for normality we will use normal Q-Q plots, filtering for both the Metro and Country areas.

```
data_metro <- data %>% filter(STAT_DIV_NAME == "Metro")
data_country <- data %>% filter(STAT_DIV_NAME == "Country")
par(mfrow = c(1, 2))

qqPlot(data_metro$INJ_OR_FATAL, dist = "norm", ylab = 'Injuries', xlab = "Norm Quantiles", main = "Normal Q-Q Plot for Metro")
qqPlot(data_country$INJ_OR_FATAL, dist = "norm", ylab = 'Injuries', xlab = "Norm Quantiles", main = "Normal Q-Q Plot for Country")
```



From the Q-Q plots we can see the data is heavily right-skewed. However, because both sample sizes are larger than $n = 30$

- Metro $n = 135944$
- Country $n = 54965$

According to the Central Limit Theorem (CLT), the sampling distribution of the mean will be approximately normally distributed when the sampling size is larger than 30. Thus, we can confirm the assumption of normality.

Assumptions For Hypothesis Testing cont.

Equal variance

To check for equal variance in the data, we will use the Levene’s test

$$H_0 : \sigma_1^2 = \sigma_2^2$$

$$H_A : \sigma_1^2 \neq \sigma_2^2$$

Where σ_1^2 and σ_2^2 are the population variance for Metro and Country accidents.

```
test <- leveneTest(INJ_OR_FATAL ~ STAT_DIV_NAME, data = data)
knitr::kable(test)
```

	Df	F value	Pr(>F)
group	1	637.8487	0
	190907		

```
test$`Pr(>F)`
```

##	[1]	1.672644e-140	NA
----	-----	---------------	----

The p -value for the Levene’s was $p < .001$, which is less than our significance level of $p = 0.05$. Therefore, we reject the null hypothesis H_0 , and determine it is *not safe* to assume equal variance.

Hypothesis Testing (Two-sample T-test)

Assuming an unequal variance we can now conduct a two-sample t-test

```
t_test <- t.test(INJ_OR_FATAL ~ STAT_DIV_NAME, data = data, var.equal = FALSE, alternative = "two.sided")
t_test
```

```
##
## Welch Two Sample t-test
##
## data: INJ_OR_FATAL by STAT_DIV_NAME
## t = -22.489, df = 81531, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Metro and group Country is not equal to 0
## 95 percent confidence interval:
## -0.1059691 -0.0889785
## sample estimates:
## mean in group Metro mean in group Country
## 1.278092 1.375566
```

Confidence Interval Test

From the Welch Two Sample t-test we can see that the 95% confidence interval for the difference in means is [-0.106, -0.089].

Because a mean difference of 0 was not captured, we reject the null hypothesis H_0 .

Analysis

The results from the Welch two-sample t-test assuming unequal variance found a statistically significant difference between the mean persons injured or killed for crashes occurring on Metro and Country roads, $t(df = 81531) = -22.489, p < .001$, 95% confidence interval for the difference in means [-0.106, -0.089]. This investigation suggests that accidents occurring on Country roads have a significantly higher number of injuries or fatalities than accidents on Metro roads.

Categorical association

Here we will determine if there is an association between the time of day and the severity of injury sustained in the crash.

H_0 : **There is no association in the population between time of day and crash severity (independence).**

H_A : **There is an association in the population between time of day and crash severity (dependence).**

```
chi2 <- chisq.test(table(data$LIGHT_CONDITION, data$SEVERITY))
chi2
```

```
##
##  Pearson's Chi-squared test
##
## data:  table(data$LIGHT_CONDITION, data$SEVERITY)
## X-squared = 713.55, df = 2, p-value < 2.2e-16
```

```
knitr::kable(chi2$observed)
```

	Fatal accident	Other injury accident	Serious injury accident
Day	2164	93827	51798
Night	1101	24611	17408

```
knitr::kable(chi2$expected)
```

	Fatal accident	Other injury accident	Serious injury accident
Day	2527.545	91686.79	53574.66
Night	737.455	26751.21	15631.34

Categorical association cont.

Critical Value test

We compare the observed value of $\chi^2 = 713.55$ to a critical value χ^2_{crit} .

```
qchisq(p = .95, df = 2)

## [1] 5.991465
```

The critical value χ^2_{crit} was found to be 5.99. Thus we reject the null hypothesis H_0 as $713.55 > 5.99$

p-value test

```
chi2$p.value

## [1] 1.136169e-155
```

We find that $p < .001$, as this is less than the 0.05 level of significance, the null hypothesis H_0 is rejected.

Analysis

A Chi-square test of association was used to test for an association in the population between time of day and severity of injury. The results of the test found a statistically significant association, $\chi^2 = 713.55, p < .001$. The results of this study suggest that accidents occurring during the night are more likely to result in fatal and serious injuries when compared to accidents during the day.

Discussion

Findings

We found a statistically significant difference between the mean persons injured or killed in crashes occurring on Metro and Country roads, $t(df = 81531) = -22.489, p < .001$, 95% confidence interval for the difference in means [-0.106, -0.089]. This suggests that crashes on Country roads results in more injuries per crash on average. Country road could potentially have a higher average speed limit when compared to the Metro. In addition to this, being in remote areas may have a negative impact on the response time of emergency services, leading to more severe injuries.

We also found that statistically significant association between the time of day and the severity, $\chi^2 = 713.55, p < .001$. Crashes occurring during the night were more likely to result in a fatal and serious injury than during the day. This potentially could be caused by lack of viability at night, or people being tired from work.

Strengths

- A large data used in this report is compiled and verified through Victoria Police crash reports, it also has a large sample size of over 190,000 observations.

Limitations

- The difference found in the means between Metro and Country was only ~0.097 injuries per crash, while it is statistically significant, it is small in practical terms.
- The statistically significant findings in the report should not be inferred as causation. Other variables such as the speed limit, type of road, involvement of alcohol were not included in the report.

Future investigations

In the future, investigations could conduct analysis on this data in conjunction with Ambulance Victoria UCL Response Times, determining if there is a relationship between a locations average response time and the number of deaths. Additionally, other variables such as the speed limit or road geometry could be investigated further.

References

- Department of Transport. (2020). *Victorian Road Safety Strategy 2021-2030*. Victorian Government. https://www.tac.vic.gov.au/road-safety/working-with-victoria-police/RoadSafetyStrategy_DEC2020.pdf
- Department of Transport and Planning. (2026). *Victorian road crash data* [Dataset]. Victorian Government Data Vic (DataVic). <https://discover.data.vic.gov.au/dataset/victoria-road-crash-data/resource/5df1f373-0c90-48f5-80e1-7b2a35507134>
- Transport Accident Commission (TAC). (2026). *Road Safety Statistics*. Victorian Government. <https://www.tac.vic.gov.au/road-safety/statistics/road-safety-statistical-summary>
- Travelvictoria. (n.d.). *M1 (West Gate Freeway), Altona North*. <https://www.travelvictoria.com.au/victoria/driving/roads/>